Development and Exploratory Evaluation of a Permutation-Based Resistance Training Framework: A Two-Year N-of-1 Case Study

Author: Kush Sethia, Fair Lawn High School, NJ, USA

Abstract

Conventional resistance training splits often overemphasize major muscle groups while neglecting individual muscular subdivisions, contributing to imbalanced development, compensation patterns and training plateaus. This study aimed to develop and preliminarily evaluate a novel subdivision- and permutation-based resistance-training framework designed to organise targeted training across muscular subdivisions through structured exercise sequencing. A single trained adolescent male (aged 14 to 16 years) independently developed and iteratively refined the framework over 24 months, incorporating three primary techniques: permutational muscular subdivision sequencing, primary-complementary training, and structured intra-session exercise ordering. Training frequency averaged three sessions per week with protein intake maintained at approximately 1.3 g/kg body weight per day. Body weight increased from 63 to 77.1 kg; one-repetition maximum improvements included bench press (30 to 65kg; 117%), deadlift (70 to 130kg; 86%), squat (60 to 110kg; 83%), and standing shoulder press (30 to 50 kg; 67%). Additionally, a pre-existing bilateral bicep curl strength imbalance of approximately 2.5 kg and a bilateral imbalance between both sides in a dumbbell medium grip vertical row of 3 kg was resolved to approximately zero. These observations occurred while the participant used the framework; however, the single-participant design, adolescent maturation, self-measurement, and other uncontrolled variables mean the causation cannot be exactly established. Multi-participant controlled studies would be needed to evaluate the framework's effectiveness.

Keywords: resistance training, N-of-1 case study, adolescent training, exercise sequencing, bilateral imbalance, primary-complementary training, training framework

Introduction

Common dilemmas as a result of imbalanced or disproportionate training include postural problems, plateaus in progress, discomfort, etc (Lloyd et al. 2014). While some customized regimens curated by professional means take these into account with other explicit concerns, some might still overlook the importance of properly sequenced exercises in the short and long term (Nunes et al. 2021). Overstimulation of major muscles and/or under stimulation of supporter and secondary muscular groups may contribute to the aforementioned points, trigger compensation patterns during training sessions, hinder neuromuscular coordination, and may also limit long-term development. Some common workout sequencing plans, like Push-Pull-Legs (PPL), the Bro Split, the Arnold Split, etc, can overemphasize major muscle groups while neglecting focused attention on muscular subdivisions. Excessive emphasis on major muscle groups, like the pectoral muscle group and the back musculature, can lead to the abovementioned concerns and underdevelopment of synergist and fixator muscles (Schoenfeld, Ogborn, and Krieger 2016)

This training framework was developed to address limitations in conventional resistance training practices by emphasising targeting of muscular subdivisions through structured sequencing using permutational models. This model was iteratively refined using evidence informed adjustments in exercise selection, split structure, and intra-session sequencing of exercises. Does a structured weight training framework implementing permutational muscle subdivisional targeting and alternating primary and complementary exercises facilitate balanced muscle hypertrophy and proportional strength development in a trained teenage male over a prolonged period of time? It was hypothesized that the disciplinary implementation of this permutational framework could directly lead to observable improvements in muscular development and progressive strength gains across major and minor muscle groups that were targeted.

Methods

Ethics and Participant Status

Parents' permission and guidance was taken while conducting this experiment. Help from coaches, AI chatbots, and peer reviewed papers was used to take decisions when developing and testing the regimen. All exercises were done under the supervision of a trained adult.

This experiment was conducted by one teenage male over the period of 2 years by making small iterations to the workout routine approximately every 10 days. After making a change, it was tested for 10 days; for how it fits in the workout session and the overall routine, checking whether or not it is comparable with other exercises performed; for fatigue patterns during the workout and in the following days; if it helped increase muscular strength for that particular subdivision by comparing change in the one-repetition-maximum for the other exercise for the corresponding muscle subdivision. This process is in parallel with incremental development, particularly periodization models that use microcycles, since it allows for changes to be made and then rejected or incorporated into the whole project. Major additions like the innovative split, the Primary-Complementary Training technique and the permutational method to exercising muscular subdivisions go through multiple iterations themselves after they have been committed to be a part of the regimen. For example, the permutational training method was first applied only to the chest and back musculatures to ensure they provide an evident advantage compared to not incorporating a complex method. Only after testing this method, it was applied to other muscle groups like the biceps and triceps. Human error must still be considered due to the small-scale of this experiment and hence any technique used must not be considered a direct advantage for a trainee, rather it should be recognized as a framework that can go through amendments to adhere to individual goals and habits. Teen individual started the training experiment at an age of 14 and ended approximately 2 years later at the age of 16. Prior training experience was a factor since the individual had practiced sports such as cycling, jogging and archery before the experiment had commenced but refrained from excessive training, apart from the experiment, during it. With an average of three training days a week, not counting rest days, approximately 1.3g of protein intake per kg of body weight per day, and at least eight hours of sleep per day the trainee aimed to reduce effects of these control variables on final results.

The Split

The workout split to be used in this comprehensive routine is one of the most important fragments of this regimen as it dictates which muscles must be trained on which day of the week. It remains simple enough to be easily understandable and useable yet complex enough to have scientific intent behind each combination of muscles targeted in every workout session.

Table 1: The Split

Day	Muscles worked
1	Chest, Back
2	Shoulder, Core
3	Triceps, Biceps, Forearms
4	Plyometrics
5	Legs
6	Rest
7	Chest, Triceps
8	Shoulder, Core
9	Back, Biceps, Forearms
10	Plyometrics
11	Legs
12	Rest

Note: The sequence of the muscles trained varies on days 1, 3, 7, and 9. Forearms are always trained last on days 3 and 9.

This split can be seen as a combination of the Arnold split, which takes advantage of the fact that training a set of antagonistic muscle pairs in the same workout (complementary training) enhances recovery, performance and muscular growth (Schoenfeld, Ogborn, and Krieger 2016) and the Push-Pull-Legs (PPL) split, where muscles that work synergistically are trained together in one training session to maximize training. This split is uniquely repeated a total of 6 times in order to change the sequence in which muscular subdivisions are targeted and to modify intensity and vary the order of muscle groups worked in a training session using the Primary-Complementary Training technique.

Breaking this split down, it is evident that there are six "major" days interspersed with "minor" days. Major days (1, 3, 5, 7, 9 and 11) involve larger muscle groups and the others, except the rest days (6 and 12), are categorised as minor days where comparatively smaller muscle groups are trained. The minor days, while allowing focus on commonly neglected areas, can serve as recovery buffers between more demanding major workout sessions to prevent overexertion. Rest days are strategically placed after ***Leg*** days (5 and 11) to allow sufficient recovery of the large lower body muscles before the next training session.

Primary-Complementary Training Technique

To vary the intensity with which a muscle group is trained in a session the Primary-Complementary Training technique, designed specifically for this routine, is utilized. Primary dictates that the muscle group is trained using the "Primary" category of its exercises which are generally movements involving bilateral training and it is recommended to perform these exercises using free wights instead of cables or machines to encourage growth of stabilizer muscles (Behm et al., 2002). Complementary dictates that the muscle group is trained with the "Complementary" category of its exercises, these are typically exercises that incorporate unilateral training using machines or cables is suggested as they allow for a greater range-of-motion per repetition and generally provide uniform resistance throughout the motion.

The complementary variation of exercises allows unilateral training which aids in the restoration of muscular balance in one's body. One method to go about this would be to train the weaker side first, which diverts more energy to it than the other side, then the relatively stronger side. This method has the potential to reduce muscular imbalance over a period of several months, with progress depending on the severity of the asymmetry and individual training response (Liao et al. 2022). To prevent the muscular imbalance shifting it is recommended to change the order in which each side of the muscle was trained after three iterations (36 days) of the Split.

The muscle groups are divided into two categories, where if one category is the primarily trained, the other is trained as complementary. This is another method to prevent overexertion of the body.

Table 2: The categories

Category 1	Category 2
Chest	Back
Shoulder	Biceps
Triceps	Plyometrics
Legs	
Core	

Note: Forearms muscle group do not utilize the Primary-Complementary training technique in order to prevent over complication.

These categories allow for a contrast in training styles between major muscle pairs, like the chest and back, which allows for more focus on one of the two major muscles trained on days 1 and 3. The focus of the workout session is directed towards the muscle being trained primarily, hence the muscle is trained before the muscle being trained as complementary, if the two categories are present. If both muscle groups are to be trained under the same category, the bigger muscle group in size is trained first, followed by the smaller muscle group. The exception to this is days when the shoulders and core are trained together, here, the shoulders are always trained first.

Table 3: Updated split

Day	Muscles worked	Day	Muscles worked
1	Chest (Primary), Back (Complementary)	13	Back (Primary), Chest (Complementary)
2	Shoulder (Primary), Core (Primary)	14	Shoulder (Complementary), Core (Complementary)
3	Triceps (Primary), Biceps (Complementary), Forearms	15	Biceps (Primary), Triceps (Complementary), Forearms
4	Plyometrics (Complementary)	16	Plyometrics (Primary)
5	Legs (Primary)	17	Legs (Complementary)
6	Rest	18	Rest
7	Chest (Primary), Triceps (Primary)	19	Chest (Complementary), Triceps (Complementary)
8	Shoulder (Primary), Core (Primary)	20	Shoulder (Complementary), Core (Complementary)
9	Back (Complementary), Biceps (Complementary), Forearms	21	Back (Primary), Biceps (Primary), Forearms
10	Plyometrics (Complementary)	22	Plyometrics (Primary)
11	Legs (Primary)	23	Legs (Complementary)
12	Rest	24	Rest

Note: The order of the muscles worked in this table is significant.

Table 3 represents the split with the Primary-Complementary Training technique applied to it, hence it is doubled to allow both the categories of muscles (see Table 2) to be trained under both styles. This new split will be repeated a total of 3 times in order to allow different permutations within muscular subdivisions. Referral to any day (1 to 12) would also refer to the day 12 days ahead (its corresponding day) in Table 3 unless otherwise stated.

Exercise Selection

Selecting exercises or movements to accurately isolate and target muscular subdivisions is a complex, yet efficacious, task (Burke et al., 2024). Two exercises are selected per subdivision of any muscle group: one primary, which would be generally be a compound exercise where stabilizing muscles are utilized, the other secondary, that would typically isolate the specific subdivision being targeted. There can be two variations for each exercise, one for being trained Primarily and one for Complementary, according to the aforementioned points. At least half of the exercises for one muscle group must have variations in order to present effect from the Primary-Complementary Training technique.

Table 4: Exercises

Muscle group	Subdivision	Primary exercise		Secondary exercise	
		Primary	Complementary	Primary	Complementary
Chest (Pectoral muscles)	Upper (Clavicle head)	Inclined chest press (dumbbell)	Inclined chest press (barbell)	Upper chest fly/Scoops	---
	Middle (Sternal head)	Bench press (barbell)	Bench press (dumbbell)	Standing cable punches	Medial chest fly
	Lower (Abdominal head)	Declined chest press (barbell)	Declined chest press (dumbbell)	Lower chest fly	---
Back (Erector Spinae)	Lower (Erector spinae group)	Deadlifts	(2,2) Hyper extensions & Seated good mornings	---	---
	Lats (Latissimus dorsi)	(3,2) Close grip lat pull downs & Wide grip lat pull downs	---	Seated supinated arm cable row (both hands)	Seated supinated arm cable row (individual hand)
	Mid back (Rhomboid Major/Minor)	Mid grip vertical row (barbell)	Mid grip vertical row (dumbbell)	Mid grip horizontal row (bar)	Mid grip horizontal row (D-handle)
	Traps (Trapezius)	Shrugs	---	Wide grip horizontal row	Wide grip vertical row
Triceps Brachii	Long head	Skull crushers (barbell)	Skull crushers (dumbbell)	Overhead press (both hands)	Cable overhead press (individual hand)
	Lateral head	Close grip bench press (dumbbell)	Close grip bench press (barbell)	Pronated triceps push downs	---
	Medial head	Supinated cable push downs (bar)	Supinated cable push downs (D-handle)	Tricep kickbacks	---
Biceps Brachii	Long head	Bicep curls (barbell)	Bicep curls (dumbbell)	Drag curls	Behind the back cable curls
	Short head	Preacher close grip curls (dumbbell)	Preacher close grip curls (barbell)	Spider curls	---
	Brachialis	Pronated curls (barbell)	Pronated curls (dumbbell)	Inward hammer curls	---
Forearms	Flexors	Pronated downwards cable wrist curl	---	---	---
	Ulnar deviation	Rope ulnar deviation	---		
	Extensors	Reverse wrist curl	---		
Shoulders	Front (Anterior deltoid)	Standing shoulder press (barbell)	Seated shoulder press (dumbbell)	Chest supported front raise	---
	Lateral (Lateral deltoid)	Seated lateral raise	---	Behind the back lateral raise	---
	Rear delt (Posterior deltoid)	Face pulls	---	Reverse cable fly (both hands)	Reverse cable fly (individual hand)

Muscle group	Subdivision	Primary exercise		Secondary exercise	
		Primary	Complementary	Primary	Complementary
Legs	Quadriceps Hamstrings Gluteal muscles Tibialis anterior & Calves	Squats	---	Leg extensions	Bulgarian split squats
		Stiff leg Romanian deadlifts	---	Leg curls	Hip adduction
		Leg press	---	Hip thrusts	---
		Shin contractions (both feet)	Shin contractions (individual feet)	Standing calves push	Seated single calf push

Note: "---" under "Complementary" states that the exercise stated under "Primary" is repeated here. If "---" is present under "Primary" and "Complementary" both then it means no exercise under that category is required. Numbers in brackets separated by a comma become prominent after the Sets section, they represent number of sets for their corresponding exercise.

While exercise selection depends entirely on the trainee's comfort, mobility, and strength, Table 4 offers a foundational template with basic exercises which can be amended and expanded upon by the trainer and the athlete. Note that the selection of these exercises was developed over the course of the research to promote efficient muscle growth and prevent overexertion.

As evident from the table, for certain muscular subdivisions, exercises that are performed bilaterally are placed in the "Complementary" column and vice-versa. This arrangement is in position to allocate more energy to a more prominent muscular subdivision when the muscle group is trained as Primary. Additionally, this approach contributes to the prevention of overexertion and allows the focal muscular subdivision within a muscle group to rotate as the split progresses.

Table 5: Exercises for core and plyometrics

Muscle group	Primary	Complementary	
		Circuit training:	
		1min work, 15sec rest	1.5min work, 30sec rest
Core	Plank Crunches Russian twists Off-set deadlifts (2,2) Suitcases & Leg ups Lying leg lifts with shifting	High knees Mountain climbers Burpees Bicycles Suitcases Crunches	Plank Wall sit Left side plank Right side plank Hollow hold Plank shoulder and toe touches
Plyometrics	Clean and jerk (1,1) Basic jumps & Tuck jumps Drop jumps Skater jumps	Box jumps --- Broad jumps Step-ups	

Note: "---" under "Complementary" states that the exercise stated under "Primary" is repeated here. Order of exercises in this table is significant and remains the same throughout the split. Numbers in brackets separated by a comma become prominent after the Sets section, they represent number of sets for their corresponding exercise.

Exercises for the core and for plyometrics are arranged in a slightly different format and are represented in Table 5. Exercise permutations do not apply to these, refer Exercise permutations.

Exercise Permutations

In order to have balanced training of not only each muscle group but also each muscular subdivision within that muscle group, the sequence of training of each muscular subdivision changes. Since there are 3 major subdivisions in most muscle groups: chest, back, triceps, biceps, forearms (exercise permutations do not apply), shoulders and legs, there are 6 (3!) different ways each muscle group can be trained, not taking into account the Primary-Complementary Training technique which increases it to 12 different approaches by which a muscle group can be worked.

For certain muscular groups—the back and legs—Table 4 represents four different subdivisions. For the back musculature, the first stated subdivision, lower (Erector spinae group), is trained before the workout and after the warm-up, hence the lower subdivision is not included in the permutations. For legs, the fourth subdivision, tibialis anterior & calves, is considered extra and is not part of the permutation. When this subdivision must be trained is stated after Table 6.

Table 6: Sequences of muscle subdivisions

Day	Muscles worked	Sequence of subdivisions
1	Chest, Back	Middle, Upper, Lower Lats, Mid back, Traps
2	Shoulder, Core	Front, Lateral, Rear ---
3	Triceps, Biceps, Forearms	Long, Medial, Lateral Long, Brachialis, Short ---
4	Plyometrics	---
5	Legs	Quadriceps, Hamstrings, Glutes
6	Rest	---
7	Chest, Triceps	Upper, Lower, Middle Medial, Lateral, Long
8	Shoulder, Core	Lateral, Rear, Front ---
9	Back, Biceps, Forearms	Mid back, Traps, Lats Brachialis, Short, Long ---
10	Plyometrics	---
11	Legs	Hamstrings, Glutes, Quadriceps
12	Rest	---

Day	Muscles worked	Sequence of subdivisions
13	Chest, Back	Lower, Middle, Upper Traps, Lats, Mid back
14	Shoulder, Core	Rear, Front, Lateral ---
15	Triceps, Biceps, Forearms	Lateral, Long, Medial Short, Long, Brachialis ---
16	Plyometrics	---
17	Legs	Glutes, Quadriceps, Hamstrings
18	Rest	---
19	Chest, Triceps	Middle, Lower, Upper Long, Lateral, Medial
20	Shoulder, Core	Front, Rear, Lateral ---
21	Back, Biceps, Forearms	Lats, Traps, Mid back Long, Short, Brachialis ---
22	Plyometrics	---
23	Legs	Quadriceps, Glutes, Hamstrings
24	Rest	---

Day	Muscles worked	Sequence of subdivisions
25	Chest, Back	Lower, Upper. Middle Traps, Mid back, Lats
26	Shoulder, Core	Rear, Lateral, Front ---
27	Triceps, Biceps, Forearms	Lateral, Medial, Long Short, Brachialis, Long ---
28	Plyometrics	---
29	Legs	Glutes, Hamstrings, Quadriceps
30	Rest	---
31	Chest, Triceps	Upper, Middle, Lower Medial, Long, Lateral

32	Shoulder, Core	Lateral, Front, Rear ---
33	Back, Biceps, Forearms	Mid back, Lats, Traps Brachialis, Long, Short ---
34	Plyometrics	---
35	Legs	Hamstrings, Quadriceps, Glutes
36	Rest	---

Note: The day numbers in this table are related to the day numbers in tables 2 and 4. "---" represents that muscle group does not use exercise permutations. The order of muscle groups on one particular day is not significant to how they are trained in this table as that is dictated by the Primary-Complementary Training technique.

For the fourth subdivision, tibialis anterior & calves, of the legs muscle group, the primary exercise is performed after the exercises for the 2nd muscle subdivision in the permutation and the secondary exercise is performed after the exercises of the 3rd muscle subdivision in the permutation (at the end).

The split is repeated 3 times as there are 6 different permutations of subdivisions for the major muscle groups and each muscle group is trained twice in a split.

Complications and exceptions

For the last subdivision in an exercise permutation of muscle groups back, chest and shoulder, only carry out the primary exercise assigned to it. For the last subdivision in an exercise permutation of muscle groups legs, biceps and triceps, only perform the secondary exercises that is allocated. For the second subdivision in an exercise permutation of muscle groups biceps and triceps, only perform the primary exercise that is assigned to it. These methods are in place to prevent overtraining of a muscle group in a workout session and overexertion of the trainee.

Muscle groups core, plyometrics (a technique of training, not a muscle group) and forearms do not utilize exercise permutations since it would lead to over complication of the workout routine.

The Complete Split

Combining the Primary-Complementary Training and exercise permutations techniques presents an elaborate workout split which promotes equal and balanced training of muscular groups and muscular subdivisions as well. To construct the complete split, Table 6 is repeated twice, with all muscle groups in category 1 (refer Table 2) being trained as Primary for the first twelve days (days 1 to 12), then being trained as Complementary for the next dozen days (days 13 to 24) and will continue to alternate for all 6 weeks. All muscle groups in category 2 will trained in the reciprocal manner. This permits each permutation of a muscle group to be trained as Primary and Complementary by the end of the complete split.

Table 7: The complete split

Dozen Days 1

Day	Muscles worked	Sequence of subdivisions
1	Chest (Primary), Back (Complementary)	Middle, Upper, Lower Lats, Mid back, Traps
2	Shoulder (Primary), Core (Primary)	Front, Lateral, Rear ---
3	Triceps (Primary), Biceps (Complementary), Forearms	Long, Medial, Lateral Long, Brachialis, Short ---
4	Plyometrics (Complementary)	---
5	Legs (Primary)	Quadriceps, Hamstrings, Glutes
6	Rest	---
7	Chest (Primary), Triceps (Primary)	Upper, Lower, Middle Medial, Lateral, Long
8	Shoulder (Primary), Core (Primary)	Lateral, Rear, Front ---
9	Back (Complementary), Biceps (Complementary), Forearms	Mid back, Traps, Lats Brachialis, Short, Long ---
10	Plyometrics (Complementary)	---
11	Legs (Primary)	Hamstrings, Glutes, Quadriceps
12	Rest	---

Dozen Days 2

Day	Muscles worked	Sequence of subdivisions
13	Back (Primary), Chest (Complementary)	Traps, Lats, Mid back Lower, Middle, Upper
14	Shoulder (Complementary), Core (Complementary)	Rear, Front, Lateral ---
15	Biceps (Primary), Triceps (Complementary), Forearms	Short, Long, Brachialis Lateral, Long, Medial ---
16	Plyometrics (Primary)	---
17	Legs (Complementary)	Glutes, Quadriceps, Hamstrings
18	Rest	---
19	Chest (Complementary), Triceps (Complementary)	Middle, Lower, Upper Long, Lateral, Medial
20	Shoulder (Complementary), Core (Complementary)	Front, Rear, Lateral ---
21	Back (Primary), Biceps (Primary), Forearms	Lats, Traps, Mid back Long, Short, Brachialis ---
22	Plyometrics (Primary)	---
23	Legs (Complementary)	Quadriceps, Glutes, Hamstrings
24	Rest	---

Dozen Days 3

Day	Muscles worked	Sequence of subdivisions
25	Chest (Primary), Back (Complementary)	Lower, Upper, Middle Traps, Mid back, Lats
26	Shoulder (Primary), Core (Primary)	Rear, Lateral, Front ---
27	Triceps (Primary), Biceps (Complementary), Forearms	Lateral, Medial, Long Short, Brachialis, Long ---
28	Plyometrics (Complementary)	---
29	Legs (Primary)	Glutes, Hamstrings, Quadriceps
30	Rest	---
31	Chest (Primary), Triceps (Primary)	Upper, Middle, Lower Medial, Long, Lateral
32	Shoulder (Primary), Core (Primary)	Lateral, Front, Rear ---
33	Back (Complementary), Biceps (Complementary), Forearms	Mid back, Lats, Traps Brachialis, Long, Short ---
34	Plyometrics (Complementary)	---

Dozen Days 4

Day	Muscles worked	Sequence of subdivisions
37	Back (Primary), Chest (Complementary)	Lats, Mid back, Traps Middle, Upper, Lower
38	Shoulder (Complementary), Core (Complementary)	Front, Lateral, Rear ---
39	Biceps (Primary), Triceps (Complementary), Forearms	Long, Brachialis, Short Long, Medial, Lateral ---
40	Plyometrics (Primary)	---
41	Legs (Complementary)	Quadriceps, Hamstrings, Glutes
42	Rest	---
43	Chest (Complementary), Triceps (Complementary)	Upper, Lower, Middle Medial, Lateral, Long
44	Shoulder (Complementary), Core (Complementary)	Lateral, Rear, Front ---
45	Back (Primary), Biceps (Primary), Forearms	Mid back, Traps, Lats Brachialis, Short, Long ---

35	Legs (Primary)	Hamstrings, Quadriceps, Glutes
36	Rest	---

46	Plyometrics (Primary)	---
47	Legs (Complementary)	Hamstrings, Glutes, Quadriceps
48	Rest	---

Dozen Days 5

Day	Muscles worked	Sequence of subdivisions
49	Chest (Primary), Back (Complementary)	Lower, Middle, Upper Traps, Lats, Mid back
50	Shoulder (Primary), Core (Primary)	Rear, Front, Lateral ---
51	Triceps (Primary), Biceps (Complementary), Forearms	Lateral, Long, Medial Short, Long, Brachialis ---
52	Plyometrics (Complementary)	---
53	Legs (Primary)	Glutes, Quadriceps, Hamstrings
54	Rest	---
55	Chest (Primary), Triceps (Primary)	Middle, Lower, Upper Long, Lateral, Medial
56	Shoulder (Primary), Core (Primary)	Front, Rear, Lateral ---
57	Back (Complementary), Biceps (Complementary), Forearms	Lats, Traps, Mid back Long, Short Brachialis ---
58	Plyometrics (Complementary)	---
59	Legs (Primary)	Quadriceps, Glutes, Hamstrings
60	Rest	---

Dozen Days 6

Day	Muscles worked	Sequence of subdivisions
61	Back (Primary) Chest (Complementary),	Traps, Mid back, Lats Lower, Upper, Middle
62	Shoulder (Complementary), Core (Complementary)	Rear, Lateral, Front ---
63	Biceps (Primary), Triceps (Complementary), Forearms	Short, Brachialis, Long Lateral, Medial, Long ---
64	Plyometrics (Primary)	---
65	Legs (Complementary)	Glutes, Hamstrings, Quadriceps
66	Rest	---
67	Chest (Complementary), Triceps (Complementary)	Upper, Middle, Lower Medial, Long, Lateral
68	Shoulder (Complementary), Core (Complementary)	Lateral, Front, Rear ---
69	Back (Primary), Biceps (Primary), Forearms	Mid back, Lats, Traps Brachialis, Long, Short ---
70	Plyometrics (Primary)	---
71	Legs (Complementary)	Hamstrings, Quadriceps, Glutes
72	Rest	---

Note: The order of muscle groups stated is significant in this table. "---" to the right of any muscle group indicates that it does not make use of exercise permutations.

Table 7 represents the split as a whole with the aforementioned techniques compiled to be used most efficiently in coordination with each other. Although utilizing the split as is may yield above average results due to the sheer focus on balanced and sufficient training of muscles, and their subdivisions, further minor techniques, when incorporated, can provide an ideal training plan. Features like repetitions, sets, rest times between exercises and sets, warming up, etc need to be addressed in a thoroughly balanced workout regimen.

Sets

Sets refer to one complete instance of repetitions for one particular movement, irrespective of the number of repetitions being performed. Commonly, a rest period precedes and follows a set, with some exceptions: such as warm-up sets, cool-down sets and the initial set of the proper workout. The number of sets executed for each exercise performs a major role in a workout as it can influence the intensity and effectiveness of an exercise and the emphasis towards the targeted muscle (Bernárdez-Vázquez et al., 2022).

For this particular workout regimen, the basic rule would be 4 sets for every primary exercise and 3 sets for every secondary exercise, excluding certain exceptions. The 4 sets for every primary exercise are structured in the following manner: The first set is treated as a warm-up set as it typically is the first movement for a muscular subdivision in a workout, except during super-sets where the warm-up set can be excluded. The next set is a working set where achieving total failure of the muscular subdivision is not necessary but a sufficient about of fatigue should be induced. The final 2 sets should be treated as failure sets where near total exertion of a muscle is achieved, assistance by a "spotter", an experienced trainer who can assist in case of failure, is recommended when performing these. For secondary exercises, all sets except the first, warm-up set, are performed. A warm-up set is not required as the muscular subdivision has already been targeted and activated in the current workout.

Exceptions

For certain muscular subdivisions—lats (Latissimus dorsi) and lower (erector spinae group)—a single primary exercise would not suffice to accurately target all parts of the muscular fibre, hence two exercises, with a lower number of sets, are used. For certain muscle groups, core and plyometrics, some pairs of drills may benefit through performing a lower number of sets due to their neuromuscular intensity. To indicate this different structure of sets for these specialty muscle groups, refer to the numbers in brackets stated before the exercise in Table 4, each number corresponds to the number of sets to be performed for the respective exercise.

Super-sets

Super-set is a label used when two sets, of different exercises, are performed in successive order with little to no rest between them. These are commonly performed with two exercises that target different parts of the same muscle group in order to activate muscular fibres that are only partially fatigued by one exercise. Super-sets can also be structured using two exercises of two antagonistic muscles. This method, whilst training one muscle, allows for a deeper stretch to the complementary (opposing), thereby aiding muscle growth for both, and vice versa. In addition to this, super-sets require less amount of time and energy than performing two movements separately. It is generally further beneficial to perform a super-set including an exercise than to skip the exercise altogether (Zhang et al., 2025). However, performing two exercises separately, rather than as a super-set, may lead to better muscular growth as each movement is allocated with sufficient energy and rest. Therefore, the number of super-sets integrated in a workout must be precisely balanced. This routine aims to maintain this delicate balance.

Super-sets are used on days where muscle groups chest and back or biceps and triceps are targeted in the same workout since these are pairs of antagonistic muscles. The specific days where super-sets are included are days 1, 3, 13, 15, 25, 27, 37, 39, 49, 51, 61 and 63. Super-sets only apply to muscle groups biceps and triceps during the third day of each dozen days, exercises for the forearms muscle group are incorporated into the workout without super-sets, this is made explicit after section Exercise Order. Super-sets take place when one Primary and one Complementary exercise have to be done for two muscular subdivisions, from two of the mentioned muscle groups, in successive order. This requirement also becomes prominent after Exercise Order.

When the conditions for a super-set are meet, it begins with four standard sets for the primary exercise of the primary muscle subdivision (the subdivision trained first, with greater focus, being trained Primarily). This is followed by three super-sets with the secondary exercises of both the primary and secondary muscle subdivisions (the

subdivision trained afterwards, typically trained as Complementary). Prior to the addition of super-sets in this training plan, the secondary exercise for the secondary subdivision being targeted would be omitted due to lack of energy being transferred to later exercises, hence the inclusion of super-sets offers a more balanced workout. After carrying out the super-sets, three sets for the primary exercise of the secondary subdivision are performed. The warm-up set is not required for this exercise since the secondary subdivision is already activated during the super-sets.

Exercise Order

The order in which exercises take place in a workout not only establishes the focal muscle group but also determines the level of engagement to the secondary muscle group(s) and subdivisions, shaping a comprehensive and well-rounded workout (Soncin et al., 2014). The conventional method to structure the order of exercises in a workout is as follows: Performing all exercises for the focal muscle group, followed by the secondary muscle group and lastly, if present, the tertiary muscular group. This method, in addition to not allocating sufficient energy to the later muscle groups, can lead to imbalanced development (Dias et al., 2010). Furthermore, it does not inherently support the inclusion of advanced training techniques like the ones mentioned above.

Pertaining to the overall goal of this workout plan, the exercise order should at least provide appropriate energy to each muscular subdivision being trained and divert major attention to the focal muscle group being trained. Additionally, it should remain simple enough in order to accommodate other, more complex, aforesaid techniques.

To start, the first (focused) subdivision of the primary muscle group is trained, followed by the first subdivision of the secondary muscle group, then lastly, if present, the first subdivision of the tertiary muscle group. This is generally repeated two more times for the total three muscular subdivisions present in most muscle groups stated in Table 4.

Exceptions

The exceptions to this are the fifth and tenth day of each week, namely, the days when the legs are targeted, these deviations have been addressed prior under the Exercise Permutations section.

Another exception to this method arises on the second and eighth day of each week: when shoulders and the core are trained in unison. During these sessions, the general method ordering principal must be used; training the shoulders first, followed by the core workout. This is due to the abnormal structure of the core exercises, which do not follow a typical subdivisional model. Instead, the core is treated as a single unit to accommodate for circuit-style training, where different cardiovascular movements are performed sequentially with fixed rest and work intervals.

Rest Periods

Appropriate rest periods, both between exercises and across the training week, are equally important to the exercises themselves. Rest intervals between sets of the same or different exercises allow the nervous system to replenish phosphocreatine stores and restores neural drive. This enables the muscles to maintain performance during successively intense movements. Insufficient rest times between sets may lead to overexertion of the muscle group, potentially impairing recovery and limiting muscular development (Schoenfeld, Pope, et al., 2016). Inversely, overly long rest periods may reduce training intensity by allowing the body to return to a near-resting state, lowering heart rate and promoting cooling down of the trained muscles (Singer et al., 2024). Due to this the rest periods must maintain a balance between. Since progressive overload (refer Repetition Range) is utilized within exercises, rest periods must follow a similar pattern. As aforementioned in Sets, usually, primary exercises consist of four sets, with each set further intense from the last one, hence, the rest periods must also start low and increase as the sets go on for one exercise. Rest intervals vary widely between individuals as the rate of central nervous system recovery depends on many factors like training experience, nutrition, sleep, lifestyle, etc (Grgic et al. 2018; Singer et al. 2024) Therefore, each trainee must approximate their preferred rest time between the sets making sure of two key points: rest duration increases as the difficulty of sets increases and the rest periods must be measured using a stopwatch-like device to minimize human error. The same values may also be used for secondary exercises, which usually consist of 3 sets.

Repetition Range

The number of repetitions performed in one set depends on multiple factors, including but not limited to the intensity of the set, its type (auxiliary or load-bearing), and its position within the workout ("Progression models in resistance training", 2009). Therefore, the exact number of repetitions in a set cannot be standardized; instead, a repetition range can help regulate it. There are several techniques that help vary the intensity of sets, particularly in accordance to progressive overload: using the same load and increasing repetitions, increasing the load to decrease repetitions at point of failure, etc. This routine, is incorporated with the latter technique as progressively increasing the load naturally reduces the number of achievable repetitions across sets in a particular exercise, aligning with the declining energy levels, and promotes heavier lifts overtime.

As aforementioned, usually, primary exercises consist of four sets, with the first set being the warm-up set and the rest three functioning as working sets. For the first, warm-up, set the repetition range to aim for is 15-17 repetitions, using a load where muscular failure is reached at approximately 20 repetitions. This helps reinforce the neuromuscular connection in subsequent sets of the exercise as the body familiarizes itself with the correct movement pattern for that exercise (Paravlic et al., 2024). The next set must be done with a load where muscular failure is induced at around 17 repetitions, while making sure 15 repetitions are performed. This set, while not acting as a set where near total failure is achieved, acts as a transition between the warm-up set and the more intense sets. It further enhances the neuromuscular coordination for the following sets. Building of this connection helps the body maintain almost proper form in intense, failure achieving, sets and reduces risk of injuries. The next, third, set is done with a repetition range of 12-15, using a weight where failure is achieved within the same range. This acts as a true failure set which significantly contributes to muscular hypertrophy (Refalo et al., 2024). The last set is also considered a failure set where the load used is such that only 7-10 repetitions can be performed, in near-perfect form, helping induce near failure of the trained muscular subdivision. This same repetition range pattern should be used for exercises with only 3 sets, only excluding the warm-up set.

If at least 12 repetitions are achieved by the trainee using load where they generally only reach 7-10 repetitions, an additional set must be performed where the athlete performs a set with a sequentially greater weight, aiming to reach complete fatigue within the 7-10 repetition range. This, new, higher weight must be considered as the standard working weight for the last set in all subsequent sessions. The help of a "spotter" (refer Sets) is recommended for this new set and all other true failure sets to reduce the risk of injury.

Drop sets

If the minimum of seven repetitions is not achieved in the last set of an exercise, one of two techniques can be used to achieve near complete failure of the muscle fibres: performing a drop set, or performing another set with a lighter load. It is seemingly common to not achieve the targeted repetitions in a set, either because of fatigue from past exercises or training sessions or psychological causes (Grgic et al. 2022). If the athlete predicts that they would be unable to perform the last, true failure set, they must, with the presence of a "spotter", perform the second-to-last set with the aim of performing seven repetitions. When at least seven repetitions for the last set are not achieved, if possible, a drop set must be performed by immediately switching to a sequentially lower weight and performing repetitions to failure. Repeating this once more would yield to a successful drop set. If a drop set is not possible, another set, with a sequentially lower weight, must be performed after an appropriate period of rest.

Training adjuncts

Warm-up

Warming-up your body prepares the body in multiple ways for the intense workout that follows and is essential to every session. It lubricates the joints by the release of synovial fluid, it raises muscle temperature, activates the nervous system, and also aids to injury prevention. Warm-up exercises must cause sweat to be produced, essentially preparing the body for the workout and setting the temperature of the muscles; they must incorporate active movements involving joints that will experience strain in the subsequent workout session; they must act

as a transitional phase between resting heart rate and high heart rate situations that are to come during the workout (McGowan et al. 2015; Ding et al. 2022).

Including three to five minutes of any particular cardiovascular exercise at low to moderate intensity in the warm-up routine can help the body reach a moderate heart rate, lubricate commonly used joints (depending on the activity) and increase neuromuscular readiness (Sople and Wilcox, 2025). Exercises like the jump rope and stationary cycling are recommended since they can easily be performed at a low to moderate intensity. They also promote lubrication in the wrist, shoulder, hip, knee and ankle joints which are commonly incorporated in high intensity compound exercises, this not only aids in the prevention of injuries, but also reduces joint stiffness allowing for greater range of motion and stronger execution of lifts (McGowan et al. 2015).

Simple active movements, including calisthenic exercises at a low intensity, focused on the musculature and joints being trained on the particular session should follow the cardiovascular drill. Arm revolutions for the shoulders (muscle and joint), wrist rotations for the wrist joint and forearm muscles, and leg swings for the hip joint are some examples of such active movements. These kinds of exercises are preferred during the warm-up since they are dynamic rather than passive, meaning that the muscles continually contract and relax through a functional range of motion instead of maintaining a static position. This active nature enhances neuromuscular activation, increases joint lubrication, and better prepares the body for the demands of high-intensity weight training (Sople & Wilcox, 2025). Passive drills, while not recommended as the only source of warm-up for a muscle group, can be included alongside active movements to further enhance mobility and reduce muscular stiffness, provided they are brief (10-15 seconds) and performed after sufficient dynamic movements. Active, and passive movements if applicable, must be performed for each muscle group involved in the following workout session in the order of priority of the muscle groups, which generally means, muscles being trained Primarily first then the muscles being trained as Complementary.

Cool-down

Cool-down acts a transitional phase between the moderate heart rate during exercises to resting heart rate. It helps active muscles relax and the trainee to revert back to a normal breathing rate by helping to getting rid of oxygen debt. Majority of the advantages from performing cool-down movements involve enhanced recovery process like by maintaining blood circulation to trained muscles to minimize excessive stiffness during recovery (Van Hooren and Peake, 2018).

Cool-down can also be used to perform mobility drills that serve a rehabilitative or maintenance purpose for a joint. Mobility drills for joints like the hip, shoulder and wrist, which are frequently put through intense demands from compound exercises, may be performed to maintain and/or better mobility of the joint. Additionally, cool-downs must include a passive stretch for the muscle groups targeted in the workout session, in the order they were trained. To add to these, a simple cardiovascular exercise can be incorporated into this part of the workout, similarly to that in warm-up. This exercise must be performed at low intensity so as to not unduly increase heart rate and could be done before the cool down session to avoid abrupt changes in heart rate (Van Hooren and Peake 2018).

Results

The results of this study have been presented below. The difference between left-handed and right-handed bicep curls and left-handed and right-handed medium grip vertical row was measured by experimentally finding the one repetition maximum for each hand for the given exercise.

Table 8: Results

Variable	Before Experiment	After Experiment	Difference (%)
Body Weight	63	77.1	22.381
Bench press 1RM	30	65	116.6667
Deadlift 1RM	70	130	85.7143
Squat 1RM	60	110	83.3333
Standing Shoulder Press 1RM	30	50	66.6667

Difference between left-handed bicep curls and right-handed bicep curls	2.5	~0	~ -100
Difference between left-handed and right-handed medium grip vertical row	3	~0	~ -100

Discussion

Numerous studies have investigated optimal exercise selection, progressive overloading strategies and training variables for muscular hypertrophy. However, much of existing literature on this topic focuses on these components in isolation, without much consideration of how they work together with other methods that may be used in a workout routine (Krzysztofik et al., 2019). The present paper aims to achieve a synthesis of several well-established principles from prior research and integrates them into a unified training program. Additionally, it introduces some new techniques not currently well-documented, for example the Primary-Complementary Training technique and the permutational sequencing of muscular subdivisions.

A primary limitation of this study is that the training framework was developed and evaluated through independent research and observation through iterative refinement by a sole trainee rather than through a controlled experiment designed to incorporate multiple participants, hence the generalizability of the proposed system to a broader population cannot be assumed. Additionally, control variables like diet, sleep and stress were given consideration and efforts were made to minimize variability by roughly maintaining workout frequency and consistency throughout the development stage of the program. Nevertheless, these factors, and others like genetics, cannot be fully controlled and their influence cannot be completely neglected.

Conclusion

The final workout regimen contains a 12-day, including rest days, rotating split that is repeated a total of 6 times to adhere to the Primary-Complementary Training and the Exercise Permutation techniques used to make most major workout sessions unique. This split is to be repeated once performed completely. It regulates sets, number of repetitions, rest periods, workout adjuncts and many more aspects of a workout regimen. These components, along with proper selection of exercises, form an integrated system designed to balance muscular development across major and supporting muscles while applying active efforts to preserving agility, mobility, training intensity and promoting recovery.

Safety and Scope Statement

This framework documents a participant's training regimen and style and is not to be considered individual media or exercise advice. Every exercise must be done under adult supervision for minor trainees to prevent injury. No exercises to take place if severe pain or injury occurs.

Author Contribution

The author solely developed and performed the training regimen under guidance and supervision of a trained adult at all times. Records were kept using the applications MyFitnessPal and Google Keep. Assistance of mentioned references and AI chatbots like ChatGPT was used when incorporating new techniques into the routine, for exercise selection and general advice.

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